PennyWise PRD (2026-01-18)

Part 1: Basic Information

Title

PennyWise

Terminology

Term	Definition
Budget	A spending limit set for a specific category over a time period
Transaction	A single income or expense entry
Category	A classification for transactions (e.g., Food, Transport)
Savings Goal	A target amount to save by a specific date
Remaining Budget	The amount left to spend in a budget category
Overspending	When expenses exceed the set budget for a category
Income	Money received (salary, gifts, refunds, etc.)

Project Information

Description

PennyWise is a personal budgeting web application designed for users who want to track their income, expenses, and savings goals. The app provides a simple, intuitive interface with visual analytics to help users manage their finances effectively without complex financial jargon.

Goals

1. Help users create and maintain monthly budgets with ease
2. Provide clear visual insights into spending patterns and trends
3. Enable users to set and track progress toward savings goals
4. Deliver timely alerts to prevent overspending

User Types

- **User (Individual)**: Can create personal account, set monthly budgets, track income and expenses, categorize transactions, view spending analytics, set savings goals, receive budget alerts
- **Admin**: Full system management, user management, view platform analytics, manage expense categories, handle support tickets

User Relationships

- Users are independent (no relationships between users)
- Admin manages all Users (1:N relationship)

Project Type

- Web Application - React
- Admin Dashboard - React

System Modules (Step-by-step Flows)

Module 1 - Budget Creation

1. User navigates to "Budgets" section
2. User clicks "Create New Budget"
3. System displays category selection and amount input form
4. User selects category and enters monthly budget amount
5. System saves budget and displays confirmation
6. Budget appears in user's dashboard with progress bar

Module 2 - Transaction Logging

1. User clicks "Add Transaction" button
2. System displays transaction form (amount, category, date, note)
3. User enters transaction details
4. System auto-suggests category based on description (optional)
5. User confirms and submits
6. System updates budget remaining and displays updated analytics

Module 3 - Savings Goal

1. User navigates to "Goals" section
2. User clicks "Create Goal"
3. System displays goal form (name, target amount, target date)
4. User enters goal details
5. System calculates required monthly savings
6. User can manually add contributions to goal
7. System displays progress with visual tracker

Module 4 - Spending Analytics

1. User navigates to "Analytics" section
2. System displays current month spending breakdown by category (pie chart)
3. System shows spending trend over past 6 months (line chart)
4. User can filter by date range and category
5. System highlights overspending categories in red

3rd Party API List

- **Google OAuth**: Social login for easy account creation
- **Firebase Cloud Messaging**: Push notifications for budget alerts

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Part 2: User Application PRD

User Types: User (Individual)

1. Common

Splash Page

- Design: App logo centered with emerald green (#10B981) background
- Auto-redirect to Login page after 2 seconds

Login Page

- **Input**:
  - Email address
  - Password
- **Next Action**:
  - Validate email format
  - Validate password (minimum 8 characters)
  - On error: Display inline error message
  - On success: Redirect to Dashboard
- **Social Login**:
  - Google OAuth button

Forgot Password Page

- **Main**
  - Input: Email address
  - System sends password reset link to email
- **Reset Password Page**
  - Input: New password, Confirm password
  - Password validation rules apply
  - On success: Redirect to Login with success message

Sign Up Page

- **Input**:
  - Email address (required)
  - Password (required)
  - Display Name (required)
  - Currency preference (optional, default: USD)
  - Monthly income (optional, for budget suggestions)
- **Rules**:
  - Email must be unique in system
  - Password minimum 8 characters
  - Display Name minimum 2 characters
- **Social Signup**:
  - Google OAuth button

2. User (Individual)

2.1 Navigation Menu

1. Dashboard
2. Budgets
3. Transactions
4. Goals
5. Analytics

6. Settings

2.2 Page Architecture & Feature Specification

1. Dashboard Tab

Main Page

1. Monthly Overview Card
    - Total income this month
    - Total expenses this month
    - Remaining budget (income - expenses)
    - Visual progress bar showing spent percentage
2. Budget Status Section
    - List of active budgets with progress bars
    - Each budget shows: Category name, Spent/Total, Percentage
    - Overspent budgets highlighted in red
    - Click budget → Navigate to Budget Detail
3. Recent Transactions Section
    - Last 5 transactions displayed
    - Each shows: Date, Category icon, Description, Amount
    - "View All" button → Navigate to Transactions Tab
4. Savings Goals Summary
    - Active goals with progress percentages
    - Click goal → Navigate to Goal Detail

2. Budgets Tab

Main Page

1. Budget List
    - All budgets displayed as cards
    - Each card shows: Category, Budget amount, Spent amount, Progress bar
    - Color coding: Green (under 80%), Yellow (80-100%), Red (over 100%)
2. Create Budget Button
    - Floating action button at bottom right

Create Budget Modal

- Category dropdown (select from existing categories)
- Budget amount input (numeric, currency formatted)
- Period selector (Monthly/Weekly)
- Save button / Cancel button

Budget Detail Page

- Budget header: Category name, Amount, Period
- Progress visualization (circular or bar)
- Transaction list for this category

- Edit Budget button
- Delete Budget button (with confirmation)

3. Transactions Tab

Main Page

1. Transaction List
    - Grouped by date
    - Each transaction shows: Category icon, Description, Amount, Time
    - Income shown in green, Expenses in red
    - Infinite scroll or pagination
2. Filter Options
    - Date range picker
    - Category filter dropdown
    - Type filter (Income/Expense/All)
3. Add Transaction Button
    - Floating action button at bottom right

Add Transaction Modal

- Transaction type toggle (Income/Expense)
- Amount input (numeric, required)
- Category dropdown (required)
- Date picker (default: today)
- Description/Note input (optional)
- Save button / Cancel button

Transaction Detail/Edit Page

- View transaction details
- Edit mode toggle
- Delete transaction (with confirmation)

4. Goals Tab

Main Page

1. Goals List
    - Active goals displayed as cards
    - Each card shows: Goal name, Target amount, Current amount, Progress bar, Target date
2. Create Goal Button
    - Floating action button at bottom right

Create Goal Modal

- Goal name input (required)
- Target amount input (required)
- Target date picker (required)
- System shows: Required monthly savings to reach goal
- Save button / Cancel button

Goal Detail Page

- Goal header: Name, Target amount, Target date
- Progress visualization (progress bar or circular)
- Current amount saved
- Days remaining
- Required monthly contribution
- Contribution history list
- Add Contribution button
- Edit Goal button
- Delete Goal button (with confirmation)

Add Contribution Modal

- Amount input
- Date (default: today)
- Note (optional)
- Save button / Cancel button

5. Analytics Tab

Main Page

1. Period Selector
    - This Month / Last 3 Months / Last 6 Months / Custom range
2. Spending Breakdown (Pie Chart)
    - Shows spending by category
    - Tap segment to see category details
    - Legend with amounts and percentages
3. Spending Trend (Line Chart)
    - Monthly spending over selected period
    - Income line vs Expense line
    - Interactive tooltips on hover
4. Category Comparison (Bar Chart)
    - Budget vs Actual spending per category
    - Visual indicator for overspent categories
5. Summary Statistics
    - Average monthly spending
    - Highest spending category
    - Month-over-month change percentage

6. Settings Tab

Main Page

1. Profile Section
    - Display name (editable)
    - Email (read-only)

    - Profile picture (optional)

2. Preferences Section
    - Currency preference dropdown
    - Monthly income (for recommendations)
3. Categories Section
    - View/Edit expense categories
    - Add custom category
    - Delete custom category (system categories cannot be deleted)
4. Notifications Section
    - Budget alert toggle (on/off)
    - Weekly summary toggle (on/off)
5. Account Section
    - Change password
    - Export my data (CSV)
    - Delete account (with confirmation)
6. About Section
    - App version
    - Terms of Service link
    - Privacy Policy link

Part 3: Admin Dashboard PRD

Admin Dashboard Standard Features

All admin dashboard pages should include the following standard features unless otherwise specified.

List Page Standard Features

Feature	Description	Required
Search	Keyword search field (name, ID, email, etc.)	✅
Filters	Status / Date / Category dropdown filters	✅
Column Sorting	Click table header to sort ASC/DESC	✅
Checkbox Selection	Row checkboxes + Select All checkbox	✅
Bulk Actions	Bulk delete / Status change / Export for selected items	✅
Pagination	Page navigation + Items per page selector (10/25/50/100)	✅

Table UI Standard Features

Feature	Description	Required

Loading State	Skeleton or spinner while data loads	✅
Empty State	Message displayed when no data exists	✅
Action Column	Edit / Delete / View Detail buttons per row	✅

Detail/Edit Standard Features

Feature	Description	Required
Detail Drawer/Modal	Click row to open detail panel	✅
Edit Form	Switch to edit mode within detail view	✅
Delete Confirmation	Confirmation dialog before deletion	✅
Audit Log	Track who/when/what was modified	Optional

Data Export Standard Features

Feature	Description	Required
CSV/Excel Download	Export current filtered/searched results	✅
Date Range Selection	Period filter for export	✅

Common UI/UX Standard Features

Feature	Description	Required
Toast Notifications	Success/Error feedback messages	✅
Breadcrumb	Current location navigation	✅
Create Modal/Drawer	Form for adding new items	✅

Page Architecture & Feature Specification

Navigation Menu (Sidebar)

1. Dashboard
2. User Management
3. Category Management
4. Platform Analytics
5. Support Tickets
6. Settings

Dashboard Home Page

Statistics Cards

- Total Users (with today's signups)
- Active Users (last 30 days)
- Total Transactions logged (this month)

- Total Budget Amount tracked
- Show increase/decrease percentage compared to previous period

Period Filter

- Today / Last 7 days / Last 30 days / Custom date range

Charts

- New User Signups trend (line chart)
- Transactions by Category (bar chart)

Recent Activity

- Recently registered users list (last 10)
- Recent support tickets

User Management

User Management Page

Main Page

1. Top Area:
    - Search: Keyword search - name, email, user ID
    - Filters: Status (Active/Suspended), Date range (signup date)
    - Export button → Download user list CSV
    - Bulk Action dropdown: Suspend / Activate / Delete
2. Table Component:
    - Checkbox column (with Select All)
    - User ID
    - Display Name
    - Email
    - Status (Active/Suspended)
    - Signup Date
    - Last Login
    - Action column: View / Suspend / Delete
3. Table Features:
    - Column sorting (click header)
    - Pagination with items per page selector

Detail Drawer

- Header Info: Display name, Email, User ID, Status badge
- Account Statistics:
    - Total transactions logged
    - Active budgets count
    - Active goals count
- Account Actions:
    - Activate / Suspend account
    - Reset password (send reset email)

- Timestamps: Created at / Last login / Last modified

Category Management

Category Management Page

Main Page

1. Top Area:
    - Search: Category name search
    - Filters: Type (System/Custom), Status (Active/Inactive)
    - Create Category button → Creation Modal
2. Table Component:
    - Category ID
    - Category Name
    - Icon
    - Type (System/Custom)
    - Status (Active/Inactive)
    - Usage Count (number of transactions using this category)
    - Action column: Edit / Deactivate (Delete for custom only)
3. Table Features:
    - Column sorting (click header)
    - Pagination with items per page selector

Create Category Modal

- Category name input (required)
- Icon selector (required)
- Type: Custom (System categories are pre-defined)
- Create button / Cancel button

Edit Category Modal

- Category name input
- Icon selector
- Status toggle (Active/Inactive)
- Save button / Cancel button

Platform Analytics

Platform Analytics Page

Main Page

1. Period Selector
    - Today / Last 7 days / Last 30 days / Last 90 days / Custom range
2. User Analytics Section
    - Total registered users

- Daily active users (DAU)
- Monthly active users (MAU)
- User retention rate
- New vs Returning users chart

3. Transaction Analytics Section

    - Total transactions logged
    - Average transactions per user
    - Transactions by category breakdown (pie chart)
    - Transaction volume trend (line chart)

4. Budget Analytics Section

    - Total budget amount tracked
    - Average budget per user
    - Most common budget categories
    - Budget compliance rate (users staying within budget)

5. Export Options

    - Download analytics report (CSV/Excel)
    - Select metrics to include
    - Date range selection

Support Tickets

Support Tickets Page

Main Page

1. Top Area:

    - Search: Ticket ID, User email, Subject
    - Filters: Status (Open/In Progress/Resolved/Closed), Priority (Low/Medium/High), Date range
    - Bulk Action dropdown: Mark as Resolved / Close / Delete

2. Table Component:

    - Checkbox column (with Select All)
    - Ticket ID
    - User (name, email)
    - Subject
    - Status
    - Priority
    - Created Date
    - Last Updated
    - Action column: View / Update Status

3. Table Features:

    - Column sorting (click header)
    - Pagination with items per page selector

Ticket Detail Drawer

- Ticket Info: ID, Subject, Status, Priority, Created date
- User Info: Name, Email, User ID
- Message Thread:
  - Original message
  - Admin responses
  - Reply input field
- Actions:
  - Update status dropdown
  - Update priority dropdown
  - Close ticket button

Settings

Settings Page

System Settings

- Default currency (for new users)
- Session timeout duration

Notification Settings

- Enable/disable system-wide push notifications
- Configure notification templates

Security Settings

- Password policy configuration
- Session management

Additional Questions (Client Confirmation Required)

Required Clarifications

#	Question	Context
1	What is the maximum number of budgets a user can create?	Need to determine if there should be a limit
2	What is the maximum number of goals a user can have active?	Need to determine if there should be a limit
3	Should there be a transaction import feature (e.g., CSV import)?	Only manual entry is currently specified
4	What are the default expense categories to include?	Need specific list of system categories

Recommended Clarifications

#	Question	Context

1	Should the weekly summary email include specific content format?	Currently only toggle on/off is specified
2	What happens to user data after account deletion?	Data retention policy needed
3	Should there be recurring transaction support?	Could simplify entry for regular bills
4	Is there a need for multi-device sync notification?	Real-time sync across devices

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Feature Change Log

Version 1.0 (2026-01-18)

Change Type	Before	After	Source
Initial Creation	-	Full PRD	PennyWise_260118_155430.md

Change Details

Initial PRD Creation

- **Source Document**: PennyWise_260118_155430.md
- **Change Description**: Initial PRD created based on training project requirements

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UX Flow Improvement Suggestions

UX flow issues identified during PRD creation:

Suggestions

#	Current Flow	Issue	Suggestion	Priority
1	Add Transaction from Transactions tab only	Most frequent action buried in tab	Add quick-add transaction button on Dashboard	High
2	Budget creation requires navigating to Budgets tab	New users may not set budgets during onboarding	Add onboarding flow prompting first budget creation	High
3	Goal contribution requires navigating to Goals > Goal Detail > Add Contribution	3 steps for common savings action	Add quick-contribute from Dashboard goals summary	Medium
4	No income tracking visibility on Dashboard	Users can't quickly see income vs expense balance	Include income indicator in Monthly Overview Card	Medium

5	Analytics is separate from related sections	Context switching between Budgets and Analytics	Add mini-analytics to Budget Detail page	Low
6	Empty state after signup	New users see empty Dashboard with no guidance	Add first-time user onboarding walkthrough	High

Suggestion Details

Suggestion 1: Quick-Add Transaction on Dashboard

- **Current**: User must navigate to Transactions tab, then click Add Transaction button
- **Issue**: Transaction logging is the most frequent user action but requires tab switching
- **Suggestion**: Add floating "+" button on Dashboard that opens Add Transaction modal directly
- **Expected Benefit**: Reduce friction for daily transaction logging, increase engagement

Suggestion 2: Onboarding Budget Prompt

- **Current**: After signup, user lands on empty Dashboard with no guidance
- **Issue**: New users may not understand they need to create budgets first to use the app effectively
- **Suggestion**: After signup, prompt user to create their first budget with a guided flow: "Let's set up your first budget! Which category would you like to track?"
- **Expected Benefit**: Better first-time user experience, faster time to value

Suggestion 3: Quick Goal Contribution

- **Current**: To add money to a savings goal: Dashboard → Goals tab → Click specific goal → Click Add Contribution → Fill form
- **Issue**: 4-step process for a common action discourages regular contributions
- **Suggestion**: On Dashboard Goals Summary, add a small "+" button next to each goal for quick contribution
- **Expected Benefit**: Encourage more frequent goal contributions, improve savings behavior

Suggestion 4: Income Visibility on Dashboard

- **Current**: Monthly Overview shows expenses and remaining budget, but income details require going to Transactions
- **Issue**: Users want quick visibility into their income to understand their financial picture
- **Suggestion**: Add "Income this month" as a clearly visible metric in Monthly Overview Card
- **Expected Benefit**: Complete financial snapshot at a glance

Suggestion 5: Contextual Analytics in Budget Detail

- **Current**: To see spending trends for a budget category, user must switch to Analytics tab
- **Issue**: Context loss when checking if current spending is normal for this category
- **Suggestion**: Add a small sparkline or "Last 3 months" trend indicator on Budget Detail page
- **Expected Benefit**: Better decision-making without context switching

Suggestion 6: First-Time User Onboarding

- **Current**: New users see empty Dashboard after signup
- **Issue**: Empty states can feel overwhelming; users don't know where to start
- **Suggestion**: Implement a 3-step onboarding checklist: 1) Set your monthly income, 2) Create your first budget, 3) Log your first transaction. Show progress and celebrate completion.
- **Expected Benefit**: Higher user activation rate, clearer path to value

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